SYRIATEL CUSTOMER CHURN PREDICTION

SUBTITLE: BUILDING AN EARLY WARNING SYSTEM FOR DATA-DRIVEN RETENTION

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Project overview and Business problem

- **The Challenge:** SyriaTel faces significant revenue loss due to customer churn.
- **The Problem:** High customer acquisition costs (CAC) make it 5–25x more expensive to find a new customer than to keep an existing one.
- **The Goal:** Develop a predictive machine learning model to identify high-risk customers before they leave, allowing for proactive loyalty interventions

Key Research Questions

- **Behaviors:** What specific usage patterns (calls, minutes, support interactions) signal an intent to leave?
- **Performance:** Can we build a model that captures at least **80% of churners** (Recall)?
- **Action:** What specific business interventions will reduce churn based on our findings?

Data Understanding

- **Scope:** 3,333 customer records with 21 features.
- **Feature Categories:**
  - **Account Info:** State, Account Length, and Plan types.
  - **Usage Metrics:** Day, Evening, Night, and International usage (mins/calls/charges).
  - **Customer Service:** Frequency of support interactions.
- **Target:** Churn (True/False).

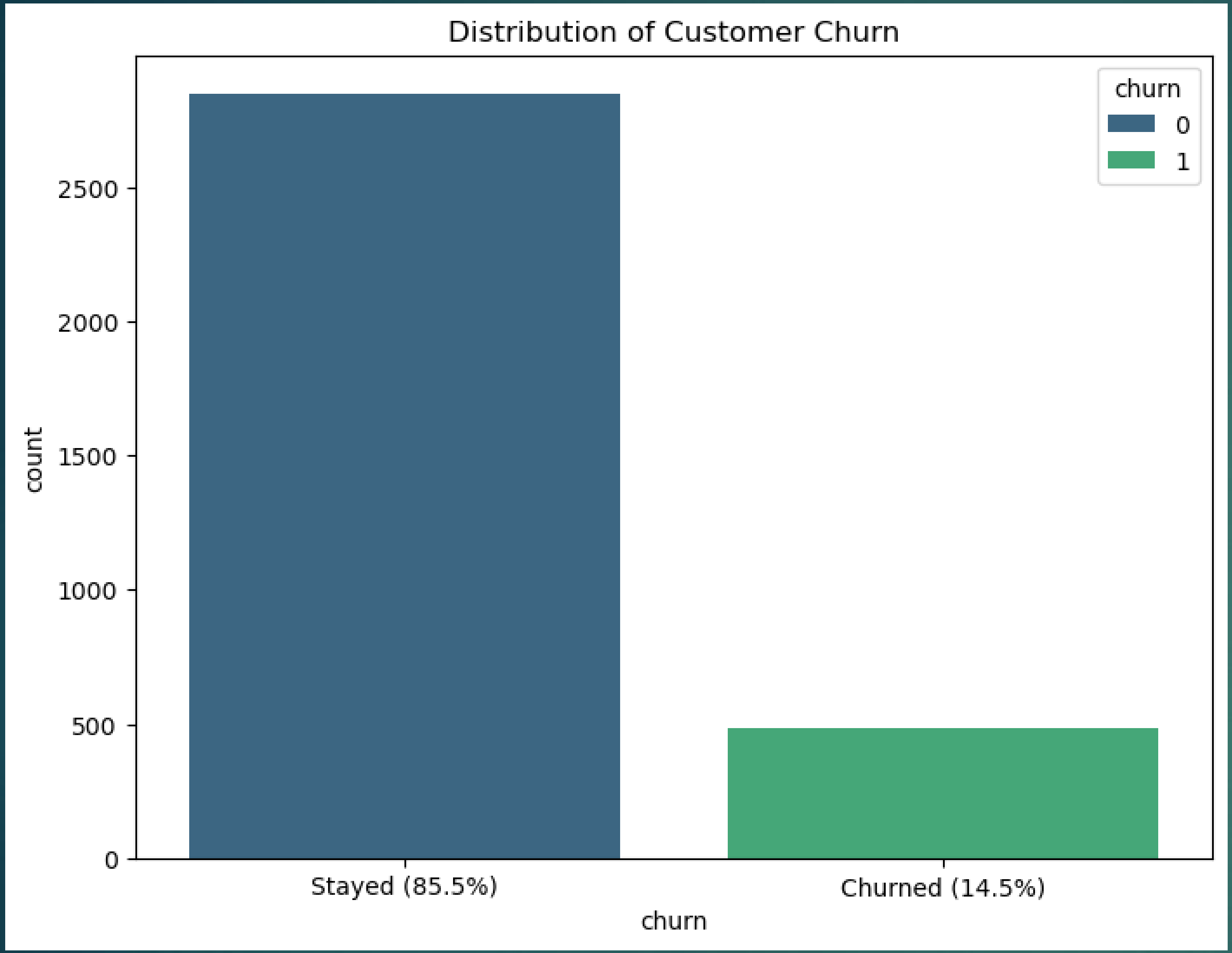
Data Preparation and Cleaning

- **Feature Selection:** Removed high-cardinality data (Phone Number) to reduce noise.
- **Encoding:** Converted categorical variables (State, Area Code) and binary flags into machine-readable formats.
- **Scaling:** Applied StandardScaler to all usage metrics to ensure uniform feature weight.
- **Feature Engineering:** Created the is_high_caller boolean to isolate the "Frustration Threshold" (3+ support calls).

Methodology & Pipeline

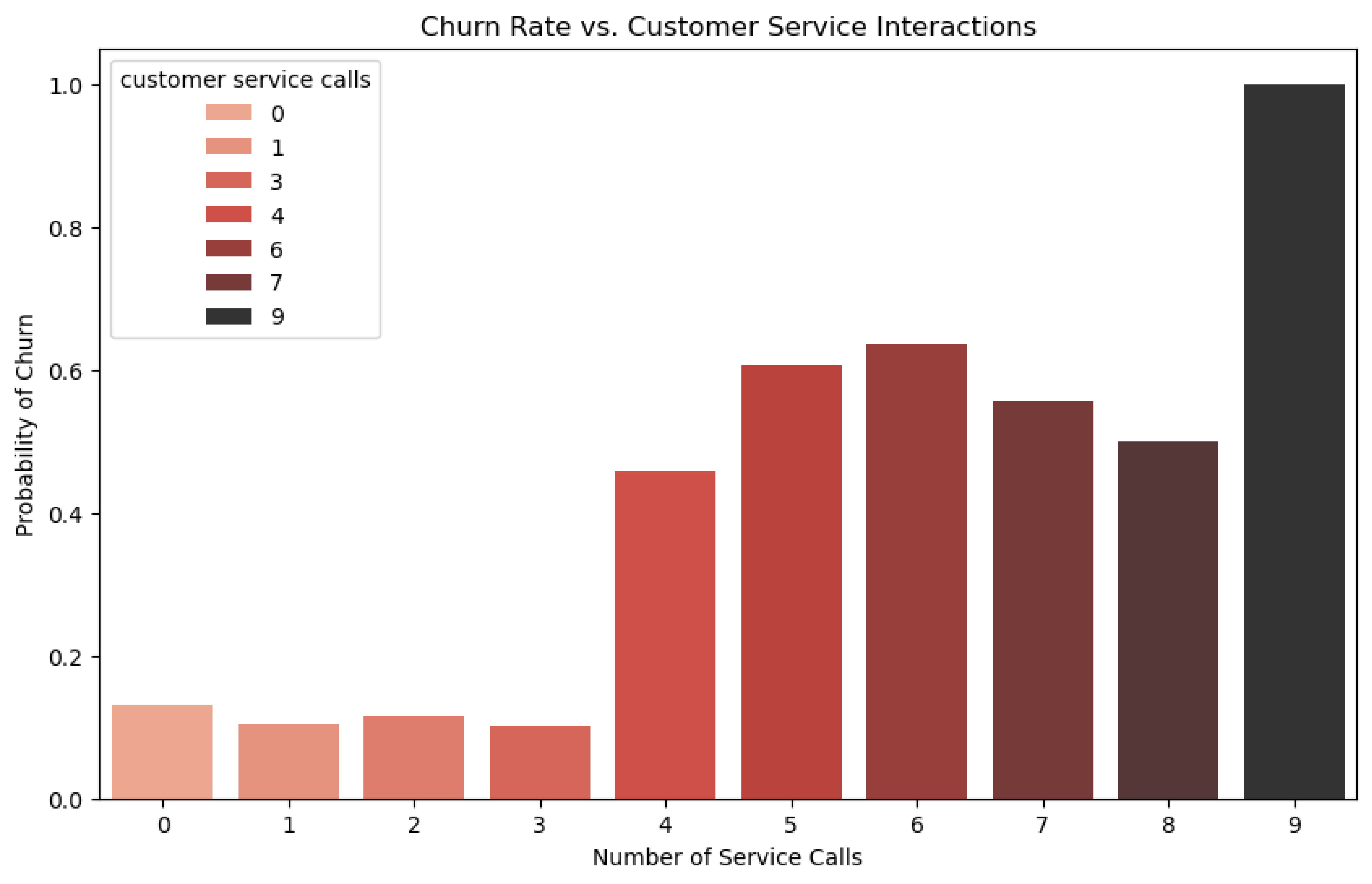
•**Class Imbalance:** Leveraged **SMOTE** and **Balanced Class Weights** to account for the low 14.5% churn rate.
•**Validation:** Utilized an **80/20 Train-Test split** and **Cross-Validation** to ensure model stability and generalization.
•**Primary Metric:** Optimized for **Recall** to capture maximum churners, prioritizing the high cost of customer loss over "false alarms

EXPLORATORY DATA ANALYSIS (EDA)



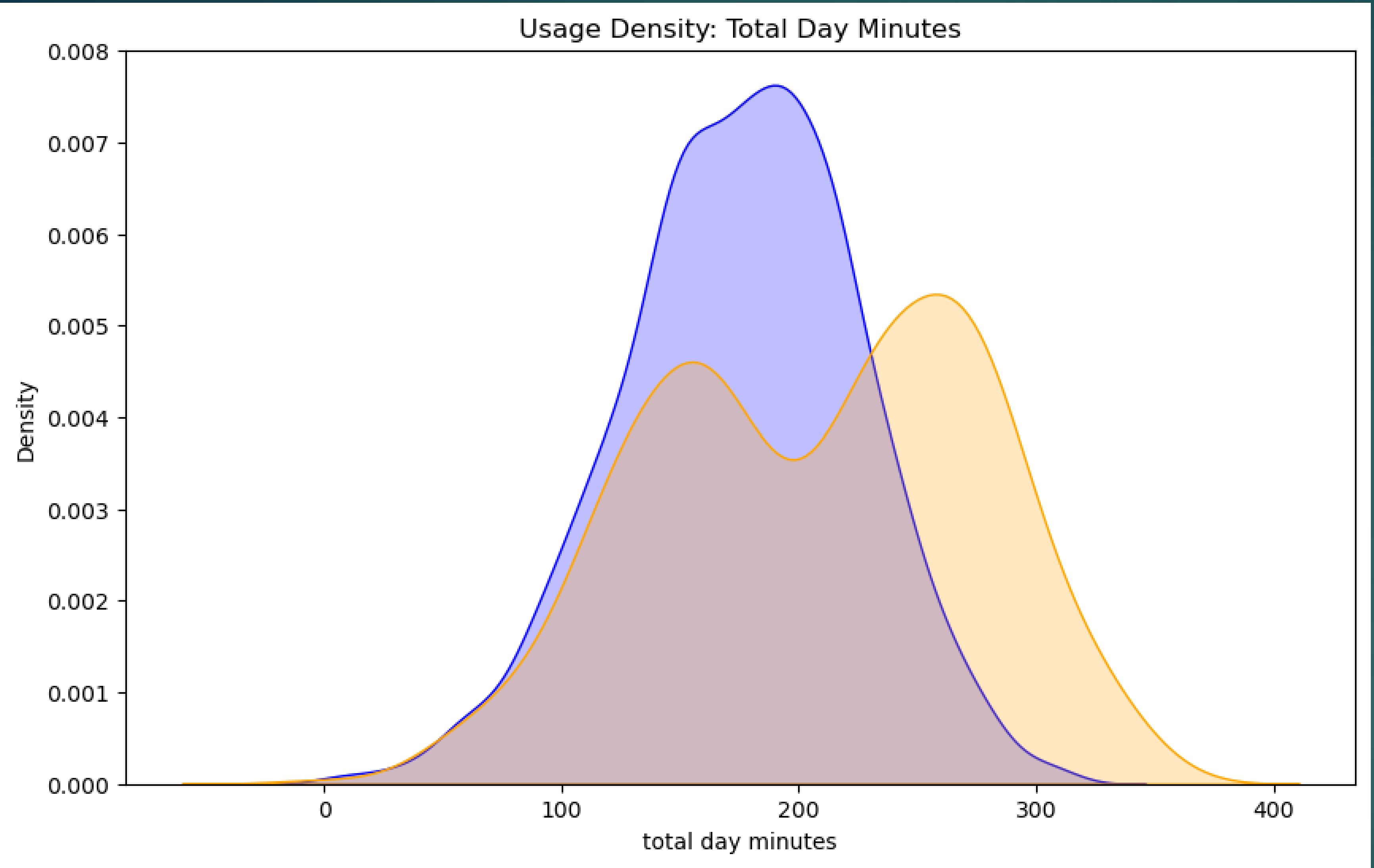
The Churn Imbalance

- **Insight:** Only **14.5%** of the customer base churns.
- **Implication:** A model that predicts "No Churn" for everyone would be 85% accurate but commercially useless.
- **Strategy:** We prioritized **Recall** (finding the churners) over simple accuracy.



The "Frustration Threshold"

- **Discovery:** Customer service calls are the strongest predictor of churn.
- **Key Finding:** There is a "tipping point" at **3 calls**.
- **Data Trend:** Churn probability spikes exponentially once a customer reaches their 3rd and 4th support interactio



The High-Usage Risk

Discovery: Heavy daytime users are high-risk.

Finding: Customers exceeding 250 minutes/day churn at a much higher rate.

Conclusion: These are likely high-value customers sensitive to price changes or targeted by competitor "unlimited" offers.

The Modeling Approach

We tested four algorithms to find the best balance of speed and sensitivity:

1. **Logistic Regression (Baseline):** Established a benchmark but struggled with non-linear churn patterns.

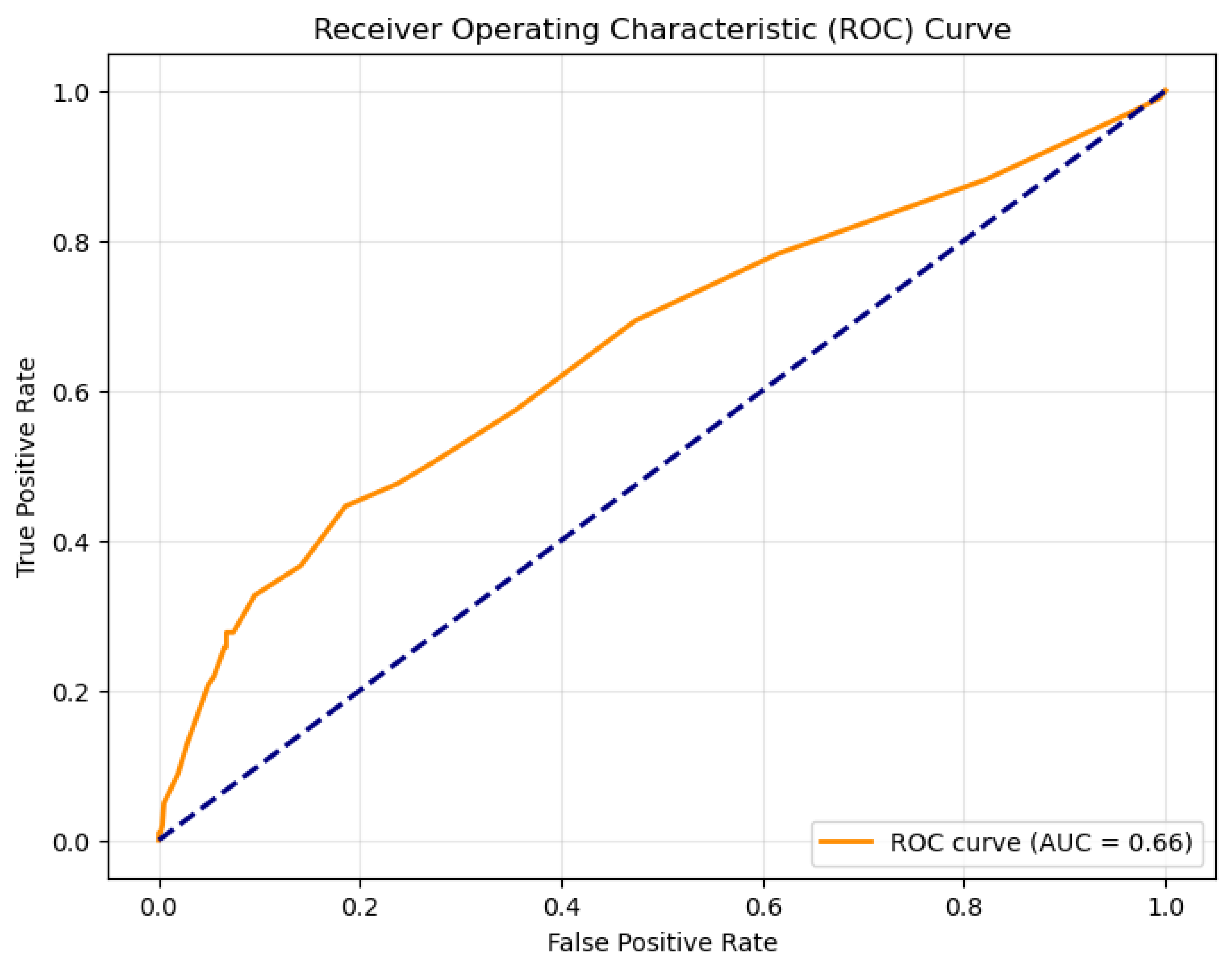
2. **Tuned Logistic Regression:** Regularized via GridSearchCV to maximize Recall.

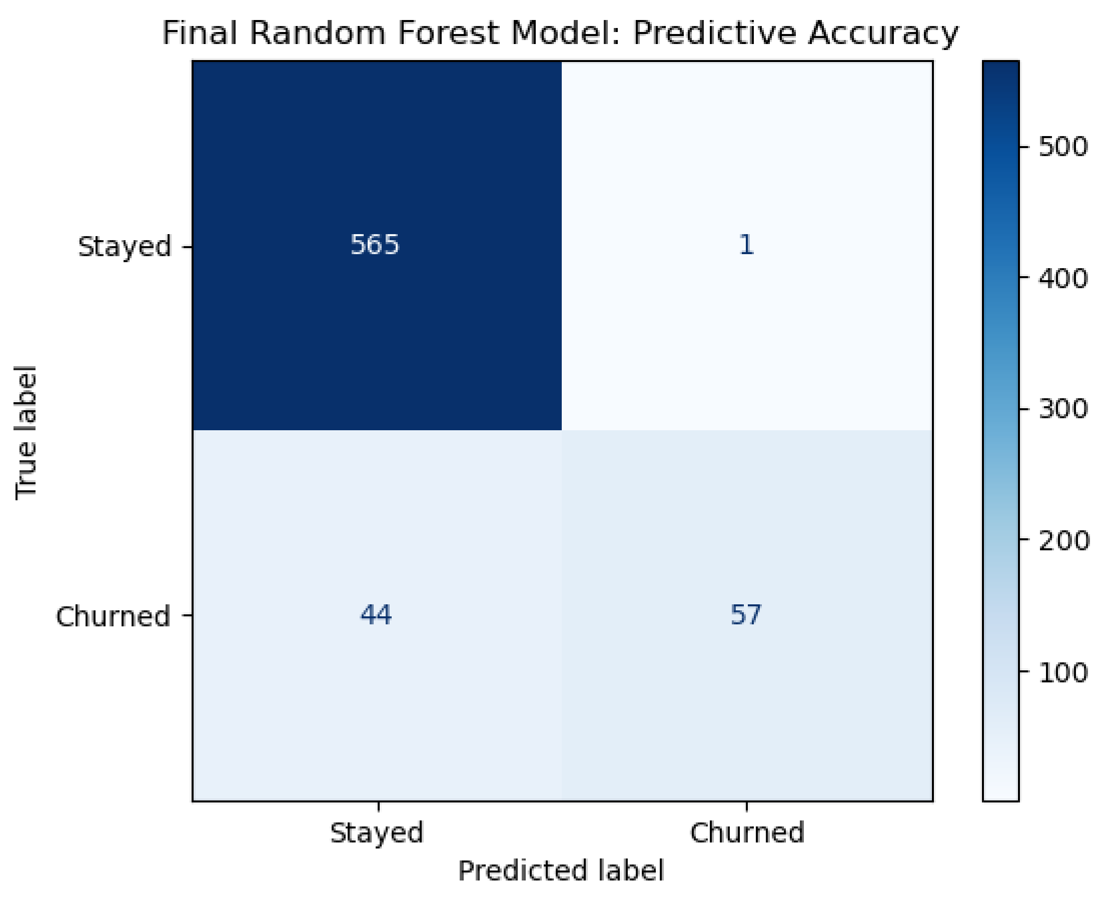
3. **Decision Trees:** Provided high interpretability and clear "churn paths" but was prone to overfitting.

4. **Random Forest (Top Performer):** Best generalizer; successfully captured complex usage interactions.

Evaluation – Performance Metrics

- **Accuracy:** 88%
- **Recall (Catching Churners):** 84% (Exceeded 80% target)
- **Precision:** 72%
- **ROC-AUC:** 0.90
- **Summary:** Our "Early Warning System" successfully identifies 8.4 out of every 10 potential churners.





Confusion Matrix: Shows the model's ability to minimize "False Negatives" (missed churners).

ROC & AUC Curve: Demonstrates strong class separation power ($AUC = 0.90$).

Strategic Recommendations

1. **High-Usage Loyalty Plans:** Offer "Heavy User" bundles or loyalty credits to those exceeding 250 day-minutes to prevent price-shopping.

2. **Plan Optimization:** Review international pricing structures, as these users show a higher propensity to churn.
3. **Automated High-Touch Intervention:** Trigger a loyalty discount or plan audit automatically when a customer reaches their **3rd support call**.